## TASK 3:

# RANDOM PASSWORD GENERATOR

**OBJECTIVE;** To Build a python tool that generators strong, random passwords based on user-defined criteria. Beginners build a command-line version; advance builds a GUI with complexity controls and clipboards integration.

**Tech Stack – Beginner:** Python, random, string **Tech Stack-Advanced:** python, secrets (cryptographically secure), PyQt5, Pyperclip ....!

Here is the code that generates a random password generator.

```
import random
import string

print ("===== Random Password Generator =====")

while True:
    try:
        # Get password length
        length = int (input ("\n Enter password length (minimum 8): "))

        if length < 8:
            print ("Error: Password length must be at least 8.")
            continue
```

```
# Character type selection
Print ("\n Choose character types:")
Print ("1. Uppercase letters")
Print ("2. Lowercase letters")
Print ("3. Numbers")
Print ("4. Symbols")

choices = input (/n "Enter your choices (example: 123): ")
 # Validate choices
if length(set(choices)) < 2:
   print ("Error: Select at least 2character types.")
   continue

if not all (c in "1234" for c in choices):
   print ("Error: Choose only 1, 2, 3 or 4.")
   continue

# Create character pool
characters = ""

if "1" in choices:
   characters += string. ascii_ uppercase

if "2" in choices:
   characters += string. Ascii _lowercase
```

```
    if "3" in choices:
        characters += string. digit

    if "4" in choices:
        characters += string. Punctuation

    # Generate password
    password = ''join (random. Choice (characters)
      for _ in range(length)
    )

    Print ("\n Generated Password:", password)

    # Generate another password
    again = input ("\n Generate another password? (y/n): ")

    if again. Lower () ! = "y":
        print ("Thank you!")
        break

except Value Error:
    print ("Error: Please enter valid numbers.")
```

**THIS COVERS THE TASK 3 CHECKLIST**

- Password length input
- Minimum **8 character**
- Uppercase letters
- Lowercase letters
- Numbers
- Symbols
- At least **2character types**
- Generates random password
- Display passwords
- Input validation
- Option to generate another password
- No need to restart the program

Actually, this is the beginner version shown in the task.

# OUTPUT;

```
===== Random Password Generator =====
Enter password length (minimum 8): 10
Choose character types:
1. Uppercase letters
2. Lowercase letters
3. Numbers
4. Symbols
Enter your choices (example: 123): 1234
Generated Password: G7@kP2#mQ9
Generate another password? (y/n): n
Thank you!
```